MOD300 Anvendt Python programmering og modellering

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Neural Network -NN-

One of the most famous models in Machine Learning is **Neural Network**.

The name comes from how the brain functions: a set of connected neurons that are either off or active.

In its essence,

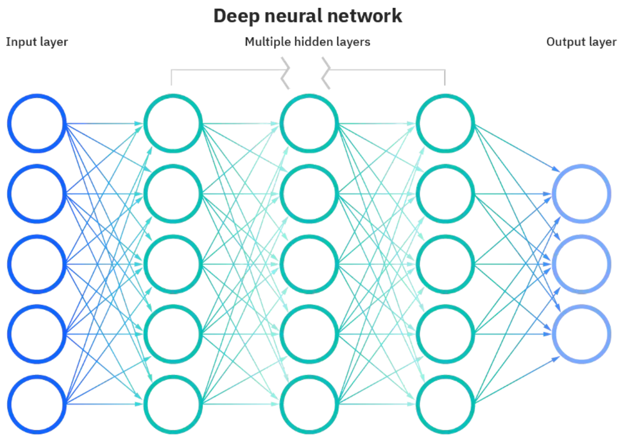
> NN is a large set of (linear) regressions executed both in parallel and in series.

Conventional NN are composed by:

1. Input layer
2. Hidden layers
3. Output layers

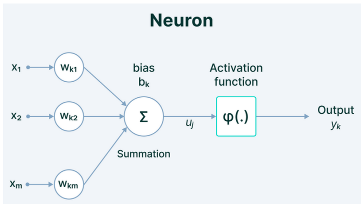
which, together, can approximately approximate any function in any dimension.

deep Neural Network



How do they work?

Each node is called neuron



Use many of these, many times, and you have builded a NN!

- It is really just a $Y = f(X)$, where w_i and b_i are the unknowns to solve for.
- As in linear regression, this becomes mainly a number of numerical recipes.
- As the problem's dimensionality is significant, several strategies have been developed.

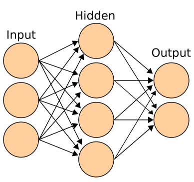
How do they work?

> 1. We thus need also an activation function.
> 2. The NN structure
> 3. Also, we might need to specify the number of epochs.

Using NN is essentially running a simulation campaign: a set of numerical experiments

Please say yes

Have you already heard of experimental design?

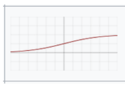


Activation function

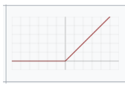
Each node applies an activation function on the weighted sum of the previous nodes.

Such functions are called activation functions. There are MANY!

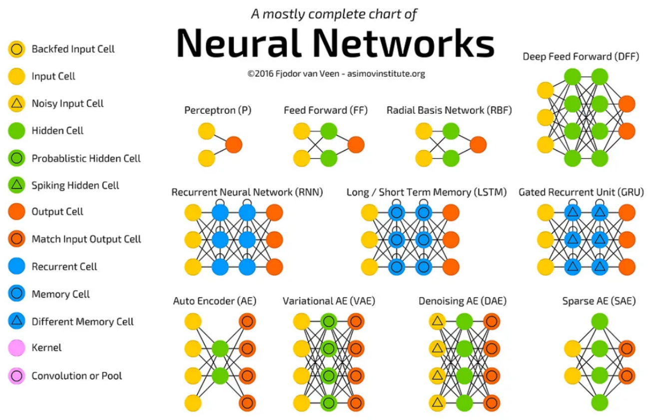
The most popular has been the logistic

	$\sigma(x) \doteq \dfrac{1}{1+e^{-x}}$	$g(x)(1-g(x))$

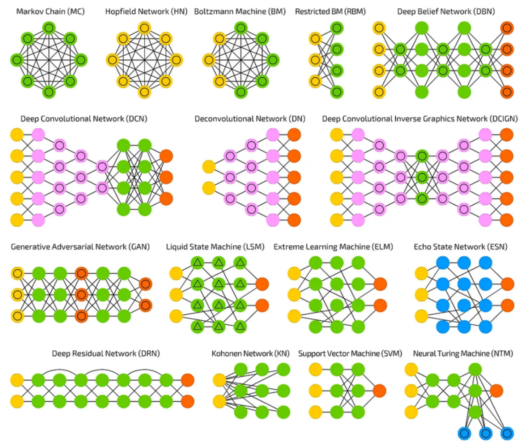
Now the most common is ReLU (Rectified linear unit)

	$(x)^+ \doteq \begin{cases} 0 & \text{if } x \le 0 \\ x & \text{if } x > 0 \end{cases} = \max(0,x) = x\mathbf{1}_{x>0}$	$\begin{cases} 0 & \text{if } x < 0 \\ 1 & \text{if } x > 0 \end{cases}$

NN Architecture



NN Architecture



NN Architecture

Neural net suffers from overfitting problems.

Even more parameters

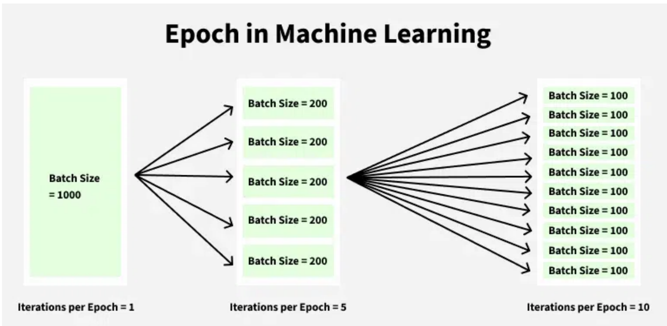
The number of nodes, the architecture, the number of hidden layer etc are NN hyperparameters

Unfortunately, at the current status of knowledge, the best architecture can be found only by trial and error.

The best fitting NN usually has a poor validation (generalizability).

NN is a very expensive approach and it should be used only if really needed.

Epoch



Epoch

Pro

1. Better performance
2. Progress tracking
3. Memory efficiency
4. Improved stopping criteria
5. More effective training

Cons

1. Overfitting risk
2. Computational cost
3. One more hyperparameter

NN types

There are many types of NN:

- ANN (artificial NN), just another name for NN
- DNN (deep) deep neural network
- RNN (recurrent NN) for audio
- CNN (convolutional NN) for images
- Autoencoder (for PCA) to compress to a latent space and decompress data
- Deep autoencoder (for interpretability)
- Physics informed NN (to merge NN to differential equations)
- ... and more ...

tensorflow, pytorch and keras are the most popular and popular libraries for NN.